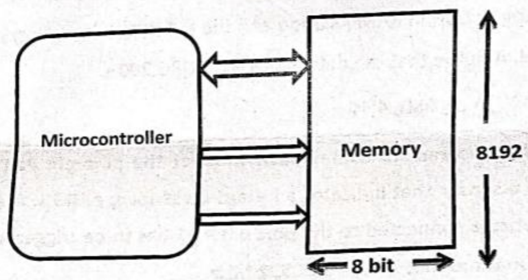
➢ DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

(10 X 10 = 100 Marks)

1. (i) Explain why microcontrollers are preferred over microprocessors for small to medium appliance applications.

   (ii) Identify the following buses given in the diagram. Also, calculate the size of address bus, data bus and the number of bits to be stored in the given memory.



2. Explain with suitable diagram the bus interface and execution unit of 8086 microprocessor.

3. Write an assembly language program to add two 8 bit data. Assume the data as 3EH and 4DH. The first data is in register R0 of bank0 and second data is in register R1 of bank 1. The result should be stored in register R2 of bank 2. Show the status of flags after the addition operation for the above data. Based on the status of each flag(set or reset) store the bit, in bit addressable RAM as follows,

Flags	Bit address(H)
Carry	7F
Aux.carry	6F
Parity	4F

4.(a) Write an 8051 assembly program to calculate y which is expressed as $y = x^3 + 9x + 7$ where x is assumed between 0 and 5 and the look-up table for x^3 has to be stored in code space from the location 300H onwards. Store x in register R0 and the final value of y in register R2.

OR

4.(b) Write an assembly language program using 8051 microcontroller instructions that finds the number is ODD or EVEN in a given data. Store the ODD data in memory location starting at 60H in RAM location and the EVEN data in memory location starting at 50H. Assume that the data is stored in ORG 200H.

DB 08H, 81H, AFH, 4EH, 68H, 4FH

5. Write an 8051-assembly program to continuously monitor the port pin P1.0 which is connected to a sensor that indicates a hazard by sending a HIGH. If a hazard is detected, a buzzer connected to the port pin P1.1 has to be triggered for 500 microseconds. Assume that XTAL = 11.0592 MHz.

6.(a) Write an assembly language program to read port P0 and transmit serially out by giving a copy to P2 do the operations simultaneously at the baud rate of 19200.

OR

6.(b) Write an 8051-assembly program that continuously gets 8-bit data from P1 and sends it to P2 while simultaneously creating a square wave of 1KHz on pin P0.1. Use Timer 1 to create the required delay for generating the square wave. Assume that XTAL = 11.0592 MHz.

7. Assume a seven segment LED display is connected in common-cathode configuration at Port1 of 8051 Microcontroller. Develop an ALP to display '0', '2' '4' with delay of 20 msec. Show the interfacing diagram.

8. Assume a switch is connected to P0.7. If SW=0, develop a program to generate a staircase waveform of 10 steps by interfacing DAC0808 with 8051 microcontroller else generate a staircase waveform with 5 steps.

9. Discuss about the dataflow model in an ARM7 with a neat sketch.

10. Explain the following instructions in ARM processor with suitable example.

    (a) CMP (b) TEQ (c) B (d) BL (e) MLA

⇔⇔⇔ Ca/D/TY⇔⇔⇔

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

(10 X 10 = 100 Marks)

1. Explain the concept of interrupt handling in an 8-bit embedded system. How does the processor manage multiple interrupts, and what is the significance of interrupt prioritization in embedded systems?

2. (i) The 8086 microprocessor has a 16-bit data bus and a 20-bit address bus. How does this affect the addressing capability, and what limitations does this impose on memory access?

   (ii) In 8086 Microprocessor, demonstrate the content of FLAGS after performing $(FFFF)_{16} + (0001)_{16}$

3. Develop an 8051 Assembly Language Program (ALP) to count the number of high bits (1s) of 8 bit data stored at RAM location 30H. Store the count in 31H.

4. For the following 8051 ALP, list the sequences of actions taking place while executing the entire program. Demonstrate your answer using proper sketch of Program Counter.

ROM LOCATION	INSTRUCTION
0000H	ORG 0000H
0001H	START:
0002H	MOV SP, #70H
0004H	MOV R0, #05
0005H	MOV R1, #00
0006H	LCALL CALC_SUM
0009H	HERE: SJMP HERE
0100H	CALC_SUM:
0101H	MOV A, R1
0102H	ADD A, R0
0103H	MOV R1,A
0104H	DJNZ R0, CALC_SUM
0106H	MOV B, R1
0107H	RET

5. Develop an 8051 ALP in which 8051 gets data from P1 and sends it to P2 continuously while incoming data from the serial port is sent to P0. Assume that XTAL=11.0592 MHz. Set the baud rate at 9600.

6. Assume there are two switches connected to MSB and LSB pins of Port 0 of 8051 Microcontroller. Develop an ALP that continuously monitors the MSB and LSB pins of P0 and based on the binary combinations; perform the actions as given in table 1.

Table1

S.No	MSB	LSB	Action
1	0	1	Generate a waveform of frequency 5KHz at P1.0 use Timer 0
2	1	0	Transmit character "A" continuously at the baud rate of 19200

7.(a) Develop an 8051 assembly code to display the string "HELLO" on a common cathode 7-segment display with a 500ms interval between each character. Assume that the pattern for "H", "E", "L", "L", "O" are stored in ROM starting at 400H. Use Timer 1 to generate 500ms of delay.

OR

7.(b) Develop an 8051 ALP to display a sequence of numbers on a 16x2 LCD. Assume the numbers (e.g., "1", "2", "3", ...till "9") are stored in ROM starting at 800H as ASCII values. The program should display these numbers sequentially on the first row of the LCD, with each number appearing for 1 second before the next number is displayed. Use Timer 0 to generate the delay.

Reference:

01 – clear display

8.(a) Draw the diagram for interfacing 8051 with DAC 0808 at Port P1. Develop 8051 assembly programs to generate (i) a stair-step ramp waveform and (ii) a triangular waveform.

OR

8.(b) Assume a pH sensor is used to measure the soil pH level and is interfaced with an 8051 microcontroller. Design the block diagram and a circuit diagram for the interfacing. The sensor provides an analog output corresponding to the pH value. Assume step size of 10mV. Explain the necessary signal conditioning and how the 8051 processes the sensor's data to display the pH level.

9. (i) The ARM core follows the RISC (Reduced Instruction Set Computing) architecture, which emphasizes delivering simple yet efficient instructions. How do the four key design principles of RISC architecture contribute to its performance in real-time applications, such as smartphones or embedded systems?

(ii) Describe the different processor modes in ARM processors and explain the corresponding register banks associated with each mode. Use diagrams to illustrate how these modes operate, highlighting how each mode facilitates efficient processing and system management.

10. (i) Explain with examples the different addressing modes for single register load-store ARM instructions.

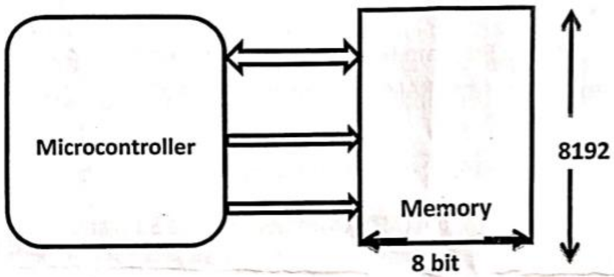
(ii) Explain the effect of barrel shifter on the following instructions in an ARM processor. What is the result of the Logical Shift Left operation on these instructions?

(a) MOV r7, r5, LSL #2

(b) ADD r0, r1, r1 LSL #1

⇔⇔⇔ W/D/TY⇔⇔⇔

1.	(i)	Explain why microcontrollers are preferred over microprocessors for small to medium appliance applications.	[5]
	(ii)	Identify the following buses given in the diagram. Also, calculate the size of address bus, data bus and the number of bits to be stored in the given memory.	[5]



2.	(i)	Explain memory segmentation in 8086 microprocessor. Explain how physical address is calculated with an example.	[5]
	(ii)	Draw the format of the 8086 Flag Register and explain the function of each bit in it.	[5]

3. Analyse the following program and specify the contents in the RAM register banks and stack pointer [SP] after execution of the following program.

```
ORG 00H
SETB PSW.3
SETB PSW.4
MOV R6, #0F5H
MOV R7, #22H
CLR PSW.4
MOV R5, #0A2H
MOV R3, #0CDH
PUSH 1EH
PUSH 0DH
```

➔ **Reset the Microcontroller at this point and execute the following program.**

```
PUSH 15H
POP 0DH
POP 17H
END
```

<u>**RAM Memory**</u>

07h		0Fh		17h		1Fh	
06h		0Eh		16h		1Eh	
05h		0Dh		15h		1Dh	
04h		0Ch		14h		1Ch	
03h		0Bh		13h		1Bh	
02h		0Ah		12h		1Ah	
01h		09h		11h		19h	
00h		08h		10h		18h	

After the Execution: SP = ?

4. Develop an 8051 Assembly Language Program (ALP) to find positive numbers from the following list and store them from RAM location 40H onwards.

   ORG 100H

   DB: 57H,66H,6FH,40H,2CH

5. Develop an 8051 Assembly Language Program (ALP) to implement the Sum and Carry expression for Full Adder circuit.

6.(a) Assume a switch is connected at P1.2. Develop an ALP for 8051 based on the status of the switch. If the SW = 0 then generate a square wave of 2KHz using timer 0 and if SW = 1, then generate a square wave of 5KHz using timer1.

OR

6.(b) Develop an 8051 ALP for the following scenario:

Imagine a sensor is connected with the pin P3.2 (INT0) to count the number of entries inside the hall. For this, you can feed the input by connecting a switch at P3.2 in which each CLOSING of switch indicates the number of incoming persons entering into the hall. Count the number of persons and display the value in Port1. When the count becomes 100 transmit the message "FULL" serially at the baud rate of 4800.

7. A seven segment LED display (common cathode) is connected to P2 of 8051 microcontroller as shown in figure below.

5 V

Write a program to display the alphabets H,E,L,L,O continuously, with a delay of 10 ms between each alphabet. Use timer0 to generate the time delay.

8.(a) With a suitable interfacing diagram, write an ALP for the 8051 microcontroller to get 8 bit digital data from ADC804. Multiply the received data by 5 and store the lower byte of the result in register R0 and higher byte R1.

OR

8.(b) Assume a switch is connected to P0.7. If SW=0, develop a program to generate a staircase waveform of 10 steps by interfacing DAC0808 with 8051 microcontroller else generate a staircase waveform with 5 steps.

9. (i) Draw the ARM core dataflow model.

(ii) Explain with a neat diagram the five stages pipeline that is incorporated in the ARM processors.

10. Determine the following using ARM instruction set to

(i) Move 8-bit immediate value into Rd.

(ii) Compare contents of Rd with 8-bit immediate value.

(iii) Add 8-bit immediate value to contents of Rd and place the result in Rd.

(iv) Subtract 8-bit immediate value from contents of Rd and place the result in Rd.

(v) Logical shift right the contents of R5 by 27 and store the result in R2.

1. In an 8-bit embedded system, how does the stack pointer and program counter interact during a subroutine call and return? What problems might arise if the stack overflows or underflows?

2. (i) Discuss the concept of "pipelining" in the 8086 microprocessor. How does the 8086 handle instruction prefetching, and what impact does this have on its performance?

   (ii) In 8086 Microprocessor, demonstrate the content of AX register and FLAGS after performing ADD AX, BX

   Given: AX = 0x8000 and BX = 0x8000

3. Develop an 8051 Assembly Language Program (ALP) to compare three 8-bit numbers stored in memory locations 30H, 31H, and 32H. Store the largest number at location 33H.

4. Analyse the following program and specify the data in the RAM locations, register A and Stack Pointer [SP] after execution of the program.

```
ORG 0
MOV SP, #60H
MOV R0, #10H
MOV R1, #20H
MOV R2, #30H
MOV R3, #40H
MOV R4, #50H
MOV R5, #60H
MOV R6, #70H
MOV R7, #80H
PUSH 02
```



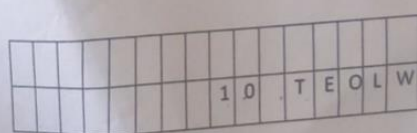
```
PUSH 03
MOV A, R1
SUBB A, R0
```

```
JZ SkipPop
POP 05
SkipPop:
PUSH 04
MOV A, R6
ADD A, R5
POP 06
ADD A, R7
END
```

5. Develop an 8051 ALP to generate a wave of frequency 2 KHz with 50% duty cycle on P1.0 using Timer 1 for the delay creation. Assume XTAL = 11.0592MHz while P1 reads data from external source which should be sent to P2 continuously.

6. Develop an 8051 microcontroller ALP in which 8051 gets incoming data from the serial port. If the received data is odd then turn ON an LED at P1.3 else sound a buzzer at P2.4.

7.(a) Develop an 8051 ALP to store the message **"WELCOME TO 8051"** starting at ROM location 0300H. Display the characters on the LCD as show in the figure below



Reference:

01 – clear display

38 – 2 lines 5X7 matrix

0E – Display ON cursor blinking

06 – Increment cursor

04 – Decrement cursor

80 – Force cursor to beginning of first line.

C0 - force cursor to beginning of Second line

OR

7.(b) Develop an 8051 assembly code to display a countdown from "3" to "0" on a common cathode 7-segment display. The program should create a 1-second interval between each digit. Use Timer 1 to generate 1-second delay.

8.(a) Assume a switch is connected to P0.7. If SW=0, develop a program to generate a ramp waveform of 10 steps by interfacing DAC0808 with 8051 microcontroller else generate a saw tooth waveform with 5 steps.

OR

8.(b) Design a block diagram and the circuit diagram to interface a Heart Rate Monitor sensor with the 8051 microcontroller. The sensor provides an analog signal corresponding to the heartbeat rate. Explain the necessary signal conditioning steps, and describe how the microcontroller processes the data to display the heart rate on an LCD.

9. How does the data flow between the ALU (Arithmetic Logic Unit) and the Barrel Shifter in the ARM processor improve performance in real-time applications, such as mobile devices or IoT systems? Explain the process with a diagram, highlighting how efficient data manipulation and shifting operations are used in real-time control systems.

10. Apply the ARM instruction set to

(i) Move an 8-bit immediate value into R7.

(ii) Add a 16-bit immediate value to the contents of R5 and store the result in R5.

(iii) Logical shift left the contents of R8 by 4 and store the result in R6.

(iv) Subtract a 16-bit immediate value from the contents of R2 and store the result in R2.

(v) Compare the contents of R10 with a 16-bit immediate value.

⇔⇔⇔ X/D/TY⇔⇔⇔

Sl. No:	Class Number(s): VL2023240505477	
1.	**What are the differences between i3, i5, and i7 processors, and how do they compare in terms of performance, features, and suitability for various computing tasks?** **ANS.**	**10**

Intel Core i3	**Intel Core i5**	**Intel Core i7**
Intel Core i3 is an entry-level processor design for low-end applications.	Intel Core i5 is a mid-range processor developed for everyday use computer systems.	Intel Core i7 is a high-end processor designed for applications that need high processing power.
Core i3 processor is a dual core processor, i.e. it has two physical cores.	Core i5 is available in quad-core (2 cores) and quad core (4 cores).	Core i7 is available in quad core (4 cores), six cores, and eight-cores.
Core i3 has a cache of size either 3 MB or 4 MB.	Core i5 has a cache of size ranging between 3 MB and 8 MB.	Core i7 has a cache of size ranging between 4 MB and 8 MB.
The clock speed of Core i3 processor is ranging between 1.30 GHz and 3.50 GHz.	The clock speed of Core i5 processor is ranging between 1.90 GHz and 3.80 GHz.	The clock speed of Core i7 processor is ranging between 2.6 GHz and 3.7 GHz.
Core i3 processor supports 4 threads.	Core i5 processor also supports 4 threads.	Core i7 processor supports 8 threads.
Core i3 processor does not support turbo boost feature.	Core i5 supports turbo boost.	Core i7 processor also supports turbo boost.
Core i3 processor does not support virtualization technology.	Core i5 processor supports virtualization technology.	Core i7 processor also supports virtualization technology.
Core i3 processor does not support Intel vPro technology.	Core i5 processor does not support Intel vPro technology.	Core i7 processor supports Intel vPro technology.
Core i3 processor is cheapest among the three.	Core i5 is mid-range processor.	Core i7 is expensive.
Core i3 processor is suitable to use for entry level tasks such as word processing, browsing, documentation, watching videos, etc.	Core i5 processor is suitable for basic work like web browsing, word processing, spreadsheets, basic graphics designing, etc.	Core i7 processor is suitable for high end applications, such as gaming, graphics designing, video editing, simulation, etc.

2.

(a) What are the key architectural components of the Intel 8086 microprocessor? Explain the function of Bus Interface Unit (BIU) with suitable diagram.

5+5 =10

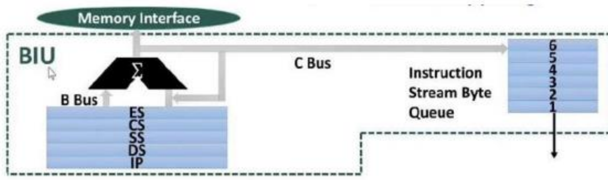
ANS.

The key architectural components of the Intel 8086 microprocessor are

Bus Interface Unit (BIU): Instruction queue, Segmentation register, Instruction pointer

Execution Unit (EU): Control Unit, Registers, ALU, Flag Register

Explain each block of BIU:



(b) Rectify errors if any and identify the addressing mode of each of the following instruction in 8086 microprocessor:

(i) MOV AX, #3FE1H

(ii) MOV DS, 3000H

(iii) MOV BL, CX

(iv) MOV AX, 50H[BX]

(v) MOV CL, [BX+SI+30H]

ANS:

(i) Error; MOV AX, 3FE1H ; Immediate addressing mode

(ii) Error; MOV AX, 3000H; Immediate addressing
MOV DS, AX; Register addressing

(iii) Error; MOV BL, [CX]; Register indirect addressing

(iv) No error; Register relative addressing

(v) No error; Base relative indexed addressing

3.

Develop an Intel 8086 assembly language program to count the odd and even numbers from a given list of 100 numbers stored at starting address of 2000:0300H and save the count values at 3000:0500H and 4000:0700H, respectively.

10

ANS:

//Comments should be written corresponding to the instructions

```
MOV AX,2000H
MOV DS,AX
MOV SI,300H
MOV CX,64H
MOV BL,00H
MOV BH,00H
```

HLT

4. **(a) Assume a data (x) is present in Accumulator. If x is negative number, write an 8051 assembly language program to transfer a string of data from code space starting at address 200H to RAM location starting at 40H in reverse order. The data is given below:** **10**

0200H: DB "Believe in yourself "

ANS:

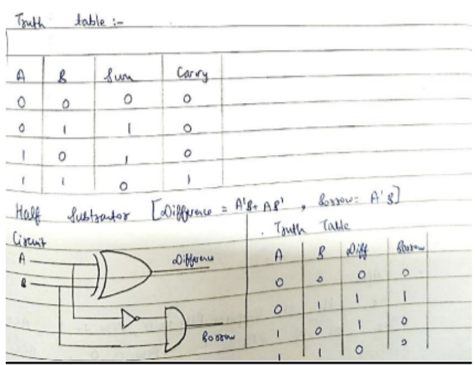
//Comments should be written corresponding to the instructions

Assume X is present in Accumulator

```
ORG 0000H
        MOV A, #0F3H; Assume any 8-bit value of X , here e.g X=0F3H is considered.
        RLC A
        JNC BYPASS
        MOV A, #00H
        MOV DPTR, #0213H
        MOV R1, #19
        MOV R0, #40H
 LOOP: CLR A
        MOVC A,@A+DPTR
        MOV @R0, A
        DEC DPL
        INC R0
        DJNZ R1, LOOP
 HERE: SJMP HERE
        ORG 0200H
        DB "VIT UNIVERSITY"
BYPASS: END
```

(b) Develop an 8051 based assembly language program to implement the expressions for half adder and half subtractor circuits. Store the sum/Difference, carry and borrow in PSW.1, PSW.5 and ACC.7, respectively. (Use Basic Logic Gates only) **10**

<u>ANS.</u>



5. **Calculate the amount of delay caused by the delay subroutine if the system has an 8051 with frequency of 16 MHz. The number of machine cycles are provided in parentheses** **10**

```
DELAY:  MOV R2, # 10010110B     (1)
AGAIN:  MOV R3, #250            (1)
HERE:   NOP                     (1)
        NOP                     (1)
        NOP                     (1)
        DJNZ R3, HERE           (2)
        DJNZ R2, AGAIN          (2)
        RET                     (2)
```

ANS:

Time period of machine cycle = 12/16MHz = 0.75 microsec

Delay for HERE loop, 5 x 250 x 0.75 =937.5 microsec

Delay for Again loop, 3 x 150 x 0.75 + 150 x 937.5 = 140962.5 microsec

Total Delay = 3 x 0.75 + 140962.5 = 140964.75 microsec

6. **Assume P1 and P2 of the 8051 are connected to LEDs and switches, respectively as shown in Figure 1. Write an 8051 assembly language program to (a) get data on switches connected to P2 and send it to the PC COM serially, (b) receive any data sent by the PC COM and put it on LEDs connected to P1. Perform both the tasks one after another continuously with 2400 baud rate.** **10**

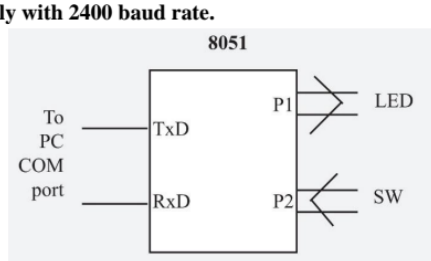


Figure 1

ANS:

7.	**Write an 8051 assembly language program that continuously get 8-bit data from P0 and sends it to P1 while the P3.3 (INT1) pin is connected to a switch which is normally high. Whenever it goes low, it should turn on an LED for 100 microsecond which is connected to P2.5 pin.**	**10**

ANS:
//Calculation of TH0 values
//Comments should be written corresponding to the instructions

```
ORG 0000H
  LJMP MAIN
;………………………ISR……………………………………………
ORG 0013H
  LJMP NEXT
```



```
  ;………………………..MAIN Program………………………..
ORG 0030H
MAIN: MOV P0,#0FFH
        MOV P1,#00H
        CLR P2.5
        MOV IE, #10000100B
HERE: MOV A,P0
        MOV P1,A
        SJMP HERE
; ………………………………………………………………..
NEXT:SETB P2.5
      MOV TMOD, #02H
      MOV TH0, #-92
      SETB TR0
L1: JNB TF0, L1;
     CLR TR0
     CLR TF0
     CLR P2.5
     RETI
     END
```

8.	**(a) Write an 8051 assembly language program to interface an 16x2 LCD by sending Hex code commands "38H, 0EH, 01H, 06H" and display "VIT Vellore" on it in line 2 position 3. Assume D0-D7, RS, R/W', and E pins of LCD are connected with P1.0-P1.7, P2.0, P2.1, and P2.2 pins of 8051 microcontroller, respectively.**	**10**

(b) Develop an assembly language program to interface a 4x4 matrix keyboard with 8051 microcontroller. **10**

ANS.

```
MOV P2, #0FFH ; make P2 input (column)
L1: MOV P1, #00H ; make P2 output (row)
     MOV A, P2 ; read all column
     ANL A, #0FH;
     CJNE A, #0FH, L2
     SJMP L1

L2: ACALL Delay ; call 20 ms delay

    MOV A, P2 ;
    ANL A, #0FH ; check if any key is pressed
    CJNE A, #0FH, ScanRow ; key pressed
    SJMP L1                 scan row
```

```
ScanRow: MOV P1, #11111110B ; ground row 0
       MOV A, P2; read all columns
       ANL A, #0FH
       CJNE A, #0FH, Row_0; key row 0, find column
       MOV P1, #11111101B ; ground row 1
       MOV A, P2
       ANL A, #0FH
       CJNE A, #0FH, Row_1; key row 1, find column
       MOV P1, #11111011B ; ground row 2
       MOV A, P2
       ANL A, #0FH
       CJNE A, #0FH, Row_2; key row 2, find column
       MOV P1, #11110111B ; ground row 2
       MOV A, P2
       ANL A, #0FH
       CJNE A, #0FH, Row_3 ; key row 3, find column
       LJMP L1
```

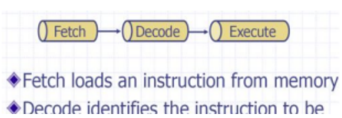
```
Row_0: MOV DPTR, #KCODE0 ; set DPTR = start of Row 0
       SJMP FIND ;         Find column key belongs to
Row_1: MOV DPTR, #KCODE1
       SJMP FIND
Row_2: MOV DPTR, #KCODE2
       SJMP FIND
Row_3: MOV DPTR, #KCODE3
       SJMP FIND

FIND: RRC A ;       Check any carry bit is low
      JNC MATCH ; if zero get ASCII code
      INC DPTR
      SJMP FIND ; keep searching

MATCH: CLR A
       MOVC A, @A+DPTR ; get ASCII from table
       MOV P0, A ; display pressed key
       LJMP L1
```

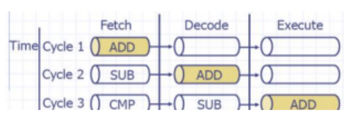
```
ORG 300H

KCODE0: DB '0','1','2','3'        ;ROW 0
KCODE1: DB '4','5','6','7'        ;ROW 1
KCODE2: DB '8','9','A','B'        ;ROW 2
KCODE3: DB 'C','D','E','F'        ;ROW 3
```

9. **Explain the 3-stage pipelining architecture of ARM7 and the 5-stage pipelining architecture of ARM9 processors with relevant diagrams depicting their pipeline structures.** **10**

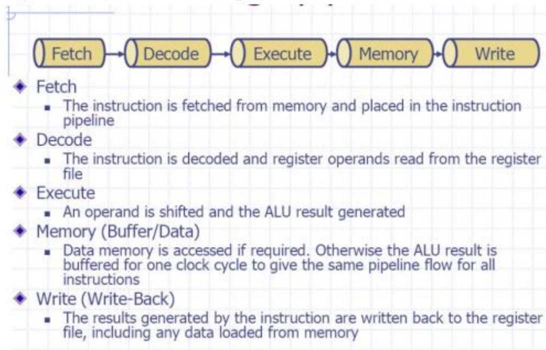
ANS:

3-stage pipelining architecture of ARM7 processor:





<table>
<tr><td></td><td>5-stage pipelining architecture of ARM9 processor:



//Explanation for both the architecture should also be included</td><td></td></tr>
<tr><td>10.</td><td>Provide a concise overview of the data processing instructions available for ARM processors.

ANS:
Overview of the data processing instructions should contain the following topics:
Data transfer instruction: Various MOV instructions; MOV instruction with Barrel shifter)
Arithmetic Instructions: ADD, SUB, MUL etc. with and without Barrel shifter
Logical Instructions: AND, OR, EX-OR etc.
Compare Instructions: CMP, TEQ, TST etc.</td><td>10</td></tr>
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1. Elaborate the working of a general-purpose 8085 or 8086 microprocessor. Provide a neatly drawn block diagram and its description.

2. Array-1 has 5 unsigned numbers and array-2 has 6 unsigned numbers. They are stored in two different data segments. Write 8086 ALP to find the smallest number in array-1 and largest number in array-2. Calculate the average of these two numbers and store it in another data segment.

3. i) Describe the internal block diagram and operations of 8255 with necessary block diagram. [5]

ii) If CS = 3000H, DS = 7000H, ES =2000H and SS = 5000H. Calculate the physical addresses corresponding to the give offset addresses. [5]

a) IP = 2000H
b) BX = 1000H
c) SI = 3000H
d) SP = 4000H
e) BP = 5000H

4.(a) Write an 8051 assembly language program to find the largest number in a given array of six numbers which are stored from external memory location 4000H onwards. Find the largest number of the array and store it at location 4017H.

OR

4.(b) Write 8051 assembly language program to convert 8-bit hexadecimal number to three-digit decimal number. Assume the hexadecimal number is stored in R0 and store the three-digit decimal number in R5, R6 and R7.

5. The text "SIMPLE INTERSET =" is present in the code (code memory) from 0100H onwards. Write 8051 ALP to calculate simple interest (SI) and transfer the text "SIMPLE INTERSET =" from the code memory to external memory from 0100H onwards. Followed by that transfer the calculated SI value to the next location of the external memory. Assume necessary input data present in data memory. (SI=PNR/100).

6.(a) Write an 8051 ALP language to generate a pulse waveforms with a duty cycle of 43% approximately (42.86%) in all the bits of the Port-1 without using timer. Don't assume the number of machine cycles for any instructions.

OR

6.(b) Write an 8051 ALP language which consists of a delay module DELAYX which can create a delay of x time units. Generate a periodic pulse waveform with frequency F = 1/(4x) and 50% duty cycle using DELAYX. Don't assume the number of machine cycles for any instruction.

7. Find the delay (D) generated by the timer-0 if it has the following settings TMOD = 01H, TH0 = 00H and TL0 = 00H. Write 8051 ALP to generate a pulse waveform with frequency f = 1/(2D) and duty cycle = 50% using timer-0 and simultaneously transfer data from Port-0 to Port-1.

Assume XTAL = 11.0592 MHz.

8. Draw the block of the described system with required control signals. A temperature sensor is connected to analog input of an ADC, the 8-bit digital output lines of the ADC are connected to P1 of 8051 and 8-bit input of the DAC is connected to P0 of 8051. Write 8051 ALP to continuously get the data (C) from the ADC output, compute F = (9*C)/5 + 32 and send the result (F) to input of the DAC.

9. Describe the ARM dataflow model and registers with block diagram and explain the operations of individual blocks.

10. Write a simple ARM program to find $y = x^2 + 3x + 5$ where x is input stored in r0 register.

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Time: **Three Hours** Max. Marks: **100**

- **KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE**
- **DON'T WRITE ANYTHING ON THE QUESTION PAPER**

Answer ALL Questions

(10 X 10 = 100 Marks)

1. Discuss the differences between microprocessors and microcontrollers, focusing on how microprocessors, like Intel Pentium and i3/i5/i7 series, are used in general-purpose computing as opposed to embedded systems.

2. Explain the internal block diagram and operational functionality of the 8255 Programmable Peripheral Interface (PPI). Illustrate your explanation with a detailed block diagram and describe how each block contributes to the overall operation of the 8255.

3. a) In an 8086 based system, given the following segment registers: CS = 4000H, DS = 6000H, ES = 3000H, SS = 8000H. Calculate the physical addresses corresponding to the following offset addresses:
   (i) IP = 1500H (ii) BX = 2500H (iii) SI = 3500H (iv) SP = 4500H (v) BP = 5500H

   b) Write an 8086-assembly language program to multiply two 16-bit unsigned integers: 1234H stored at memory location 2000:0700H and 5678H stored at 2000:0702H. Store the 32-bit result at memory locations 3000:0800H (lower 16 bits) and 3000:0802H (higher 16 bits).

4.(a) In the 8051 Microcontroller, develop an assembly program to add two 16-bit BCD numbers 4567H and 1234H. Store the result in registers R5 (lower byte) and R6 (higher byte), ensuring correct BCD carry handling.

OR

4.(b) (i) In a version of 8051, the crystal frequency is 16MHz. Find the time delay associated with loop section of the following DELAY subroutine.

Label	Instructions	No. of Machine Cycles (MC)
DELAY:	MOV R5, #250	1 MC
Here:	NOP	1MC
	NOP	1 MC
	NOP	1 MC
	NOP	1 MC
	DJNZ R5, HERE	2MC
	RET	2MC

(ii) In the 8051 microcontrollers, determine the status of the Carry (CY), Auxiliary Carry (AC), and Parity (P) flags of the Program Status Word (PSW) register after performing the binary addition of the values 88H and 93H.

5. Develop an 8051-assembly program to find all the even numbers from the list stored in code memory starting at ORG 300H: 25H, 36H, 45H, 50H, 72H. Store the even numbers in RAM starting from 60H.

6.(a) Write an 8051-assembly language program to control an LED array connected to Port 0. The system should generate a square wave on all pins of Port 0, with a 3ms ON time and a 10ms OFF time. The microcontroller operates with a 12 MHz crystal oscillator. Use any timer mode of the 8051 to configure the timer for generating the required delays.

OR

6.(b) The system based on 8051 Microcontroller continuously monitors a sensor connected to pin P1.2, detecting objects on the conveyor. When an object is detected (P1.2 = high), the microcontroller inserts a 600 μS delay (for sensor debouncing) and then activates the motor by writing 25H to port P2. Show the necessary delay calculation using XTAL = 11.0592 MHz.

7. Write an 8051 – assembly language program using interrupts to do the following:

a) Receive data serially and sent it to P0,

b) Have P1 port read and transmitted serially, and a copy given to P2,

c) Make timer 0 generate a square wave of 5kHz frequency on P0.1.

Assume that XTAL-11,0592. Set the baud rate at 4800.

8. Develop an assembly language program to interface a 4x4 matrix keyboard with 8051 microcontrollers.

9. Explain the distinct processor modes supported by ARM processors. Also, describe the Five-stage pipeline mechanism in the ARM processor, accompanied by a diagram to clarify its operation

10. a) Provide a brief overview of the data processing instructions available for ARM processors.

b) Write an ARM program to find the sum of all elements in an array. Assume the base address of the array is stored in r0, and the number of elements is stored in r1.

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(10 X 10 = 100 Marks)

1. (i) Differentiate between Microprocessor and Microcontroller with example. [5]

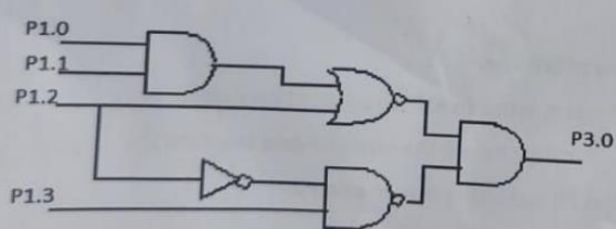
   (ii) Mention the criteria's for choosing a microcontroller for developing an embedded product. [5]

2. Explain with suitable diagram the bus interface and execution unit of 8086 processor.

3. Show the contents of various locations after executing each and every instruction of the following code.

Line No	Program	Data in RAM	
1.	MOV 28H,#4BH	28H :	; SP :
2.	MOV 0FH,#3EH	0FH :	;SP :
3.	MOV 1FH,#5CH	1FH :	; SP :
4.	MOV SP, #49H	SP :	
5.	PUSH 28H	28H :	; SP :
6.	PUSH 0FH	0FH :	; SP :
7.	PUSH 1FH	1FH :	; SP :
8.	POP 3CH	3CH :	; SP :
9.	POP 4DH	4DH :	; SP :
10.	PUSH 4DH	4DH :	; SP :
11.	POP 5FH	5FH :	; SP :
12.	END		

4. Develop an assembly code for 8051 microcontroller, to implement the digital circuit as shown in figure.



5. Develop an ALP for 8051 Microcontroller to initiate with counter 1 in mode 2. When the count reaches 25, turn on the buzzer for 1 second which is connected in port pin P1.2.

6.(a) Write an 8051 ALP to monitor the received character from serial port at 4800 baud rate. If the received character is 'L' then generate a waveform of 25% duty cycle. If the received character is 'H' then generate a waveform of 75% duty cycle at pin P1.0. The crystal frequency is 11.0592 MHz.

OR

6.(b) Write an 8051 assembly language program to get the data from Port P0 and send it to Port P1 continuously, while the interrupts will do the following, (Assume XTAL freq = 11.0592 MHz)

(i) Keep transmitting the data 'SAVE WATER' continuously through the serial COM port with 2400 baud rate.

(ii) Whenever there is a falling edge at P3.3 (INT1) buzzer at P0.4 for some time.

7. Develop an 8051 assembly program for the below LCD display [2 rows, 16 characters]. Assume "VITVELLORE" is stored in ROM starting from 200H. Bring the characters that is to be displayed in LCD from ROM location starting at 200H

											R	L	E	T	V

Reference:

01 – clear display

38 – 2 lines 5X7 matrix

0E – Display ON cursor blinking

06 – Increment cursor

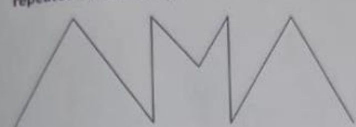
04 – Decrement cursor

80 – Force cursor to beginning of first line.

8.(a) With a suitable interfacing diagram, write an ALP for the 8051 microcontroller to get 8 bit digital data from ADC804. Square the received data and send the lower byte of the result to port P0 and higher byte to port P2.

OR

8.(b) With suitable DAC interfacing diagram with 8051 microcontroller, write a program to generate a waveform as shown below. The pattern should be repeated continuously.



9. Discuss the different processor modes of ARM 7 processors with neat sketches describing their register banks.

10. Explain the different categories of instruction set in ARM 7 with example for each category.

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